

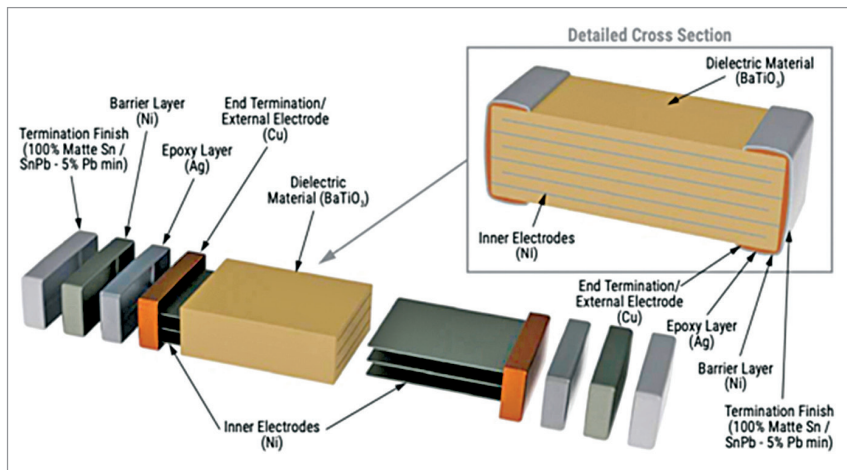


Fig. 6: Construction of FT-CAP MLCC (Credit: Kemet)

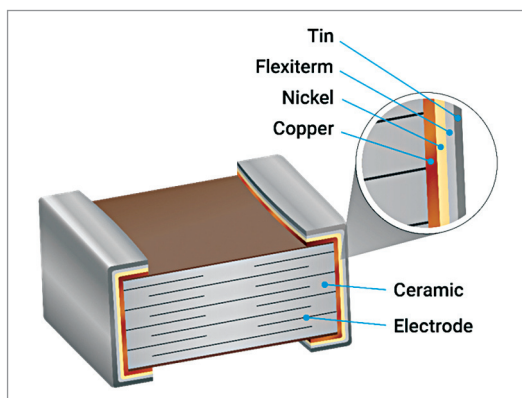


Fig. 7: The additional layer of a Flexitem MLCC (Credit: AVX)

performance with a low ESR.

Using hybrid capacitors. These comprise an anode with a liquid and conductive polymer construction, alongside a cathode made from aluminium. The large surface area of the liquid enables elevated capacitance values to be achieved in a compact form factor. Like the previous option, this comes with added costs.

The EEH-ZS hybrid series of capacitors from Panasonic particularly target automotive applications and have vibration-resistant properties. These have auxiliary side terminals that enhance their vibration resistance. This is why these capacitors can withstand shocks with vibration acceleration up to 30G. Further, the side terminals make soldering easier. While using these capacitors,

designers may choose not to deploy external anti-vibration techniques. These are also RoHS/REACH compliant and qualify the requirements set by AEC-Q200.

Using vibration-resilient MLCCs. In MLCCs, it is possible to make certain enhancements to the methods via which the capacitor body and its terminations are interconnected with one another so that the vibrational forces experienced

are dampened. In this way, additional costs of using polymer or hybrid capacitors can be avoided. This enables greater mechanical robustness to be attained, while still benefiting from the cost advantages of MLCCs.

To address the need for capacitors that are able to resist vibrational forces, Kemet developed its FT-CAP series. These MLCCs have an innovative termination arrangement that helps to lower the strain imparted from the PCB when it is flexed. A conductive silver epoxy layer is included in the termination structure (situated between the base metal and nickel barrier layers). This enables terminal strength to be upheld but provides the cushioning necessary to prevent any cracks from appearing. The

capacitor's continued performance is thus assured.

Kyocera AVX is also tackling the vibration issues previously outlined through its Flexitem low ESR MLCC products. These also have flexible terminations fitted to them for added elasticity. Here nickel-plated copper terminations are supplemented by a proprietary layer that absorbs incident mechanical stresses. The manufacturer has successfully managed to do this without there being any major increase in the costs involved. They support an -55°C to +125°C operational temperature range and 5mm flexes can be comfortably handled by them (compared to the 2mm that MLCCs are normally rated for).

A comparison of the capacitors mentioned above is given in the Table.

There are many capacitor series developed by other manufacturers to combat specific issues, not only vibration. They leverage engineering advances that provide enhanced levels of ruggedness. For example, the PCZ series introduced by Nichicon Corp. has a great degree of heat resistance.

The automotive industry places exacting demands on the electronic hardware employed in vehicle designs, and the next generation of vehicle models (particularly those that are EV or hybrid) will only exacerbate this. High voltage levels will call for capacitors that have superior performance characteristics. These components will need to withstand heavy vibrations, for instance, as they are now being placed into more difficult surroundings. This means that automotive engineers need to select their components wisely to be assured of long-term reliability even under the harshest of working conditions. **EFY**

This feature is inspired by an article authored by the technical content team at TME. The co-author, Aaryaa Padhyegurjar, is an Industry 4.0 enthusiast with a keen interest in innovation and research.